

Mars-JEPA: Multispectral Joint Embedding Predictive Architectures for Martian Landslide Segmentation

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Abstract

The expansion of autonomous exploration on Mars has unlocked unprecedented scientific frontiers, yet it requires navigating a landscape characterized by frequent landslides that drive geomorphological change by mass movement of rock and debris. This presents critical mechanical hazards to robotic assets. Mitigating these risks through accurate landslide prediction is therefore essential to safeguarding mission longevity and preventing catastrophic loss. While landslide prediction from multispectral images using traditional methods yields high accuracy, its generalizability is limited. In this work, we study the performance of self-supervised learning-based (SSL) image models, specifically trained as joint-embedding predictive architectures (JEPA) and develop a novel physics-aware image encoder, showing additional performance gains while maintaining the generalizability provided by SSL models. Experiments reveal that our physics-aware masking strategy captures long-range dependencies between spectral bands. We show that by learning cross-spectral latent correlations rather than raw pixel reconstructions, our model is significantly more robust to the sensor noise and varying lighting conditions commonly encountered in multispectral remote sensing data. We train JEPA encoders employing different masking strategies, and benchmark their downstream landslide segmentation performance on the MMLSv2 dataset. This work directly improves autonomous safety for planetary exploration and establishes a scalable framework for self-supervised learning in high-dimensional remote sensing and terrestrial disaster response.

1 Introduction

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2 Related Work

The development of a robust landslide detection system for the Martian surface requires an interdisciplinary understanding of computer vision and planetary science. This section reviews recent Section 2.2, the specific challenges of Section 2.1, and the paradigm shift toward Sections 2.3 to 2.5.

2.1 Planetary Surface Analysis and Exploration

Mars has long been a focal point for planetary exploration, with significant research dedicated to identifying landslides on its surface. Detecting landslides automatically is a non-trivial task due to shifting atmospheric conditions and complex geomorphology. Historically, experts have identified these features manually using high-resolution imagery from the High Resolution Imaging Science Experiment (HiRISE) [1].

Early automated systems primarily utilized physics-based features derived from digital elevation models, such as slope and elevation. Recently, the field has transitioned toward deep learning, employing Convolutional Neural Networks (CNNs) [2] for automated mapping. While traditional studies relied on monochromatic imagery, the introduction of the MMLSv2 dataset [3] enabled the use of multimodal data, combining spectral imagery with topographic information to improve detection accuracy.

2.2 Advancements in Semantic Segmentation

Semantic Segmentation has advanced from basic pixel-wise classifiers to encoder-decoder frameworks designed to capture multi-scale context. Early designs like U-Net became highly popular because of its simple design based on skip connections that preserved spatial features. While U-Net++ improved this by using nested and dense skip pathways to bridge the semantic gap between the features. To handle objects at different scales, Feature Pyramid Network (FPN) and Pyramidal Attention Networks (PAN) focussed on integrating high-level global semantics with low-level spatial features.

Further refinements led to the development of modules like DeepLabv3+ which relied on atrous convolutions to expand receptive fields. More recently attention driven architectures like MANet and UperNet have sought to capture global dependencies effectively. Most recently transformer based frameworks like SegFormer showed the efficiency of hierarchial self-attention in capturing long range concept.

2.3 Self-Supervised Learning in Remote Sensing

Self-supervised representation learning has seen a paradigm shift from contrastive frameworks, which rely heavily on data augmentations, to Masked Image Modeling (MIM). While generative approaches like Masked Autoencoders (MAE) [4] have shown promising results by reconstructing raw pixels, they are often ill-suited for the multi-sensor nature of Martian remote sensing. Pixel reconstruction is computationally intensive for high-dimensional data and risks "information leakage" in multimodal datasets, where highly correlated channels (e.g., RGB and Grayscale) allow the model to bypass learning deeper geological features.

2.4 Joint Embedding Predictive Architectures (JEPA)

The JEPA framework has recently been proposed as a foundational step toward non-generative alternative for developing world models [1]. Unlike generative models, JEPA performs its predictive objective entirely within an abstract representation space, capturing. Following the I-JEPA approach [2], our model utilizes an asymmetrical structure consisting of a context encoder, a target encoder, and a predictor.

The context encoder processes visible portions of an image, while the predictor is tasked with estimating the latent embeddings of missing regions. This "prediction in latent space" encourages the network to capture underlying structural and semantic properties. A significant challenge in such self-supervised frameworks is representation collapse—where the encoder converges to a trivial constant output. To maintain feature diversity, methods such as Stochastic Information Gradient Regularization (SIGReg) are often employed to ensure that learned representations remain informative across multiple input dimensions.

2.5 Spatio-Spectral Reasoning

Spatio-Spectral Reasoning involves the simultaneous modelling of spatial structures and spectral ranges across diverse sensor modalities. Traditional multimodal implementations often treat multi-channel inputs as a single unified cube; however, an emerging strategy involves independent channel tokenization paired with factorized learned embeddings. This approach enables a more granular masking strategy where specific sensors or modalities can be hidden at specific spatial locations. Such strategies theoretically force a model to learn the intrinsic physical correlations between disparate sensors—such as the relationship between topographic slope and thermal inertia—leading to more robust features for downstream tasks like landslide segmentation.

3 Methodology

In this work, we study the downstream performance on segmentation tasks for two JEPAs trained with different tokenization masking strategies, on multispectral images. This section details the formulation and development of the self-supervised models for segmentation as follows: Section 3.1 formulates the landslide segmentation problem on multispectral images; Section 3.2 provides an overview of the general architecture and methodology to train a JEPA encoder; Section 3.3 discusses the adaptation of the I-JEPA [?] architecture for multispectral images; Section 3.4 provides a novel spatial-spectral masking strategy for training JEPAs on multispectral images, and finally Section 3.5 discusses the integration of the JEPA encoders with SOTA segmentation architectures for landslide segmentation on the Martian surface.

3.1 Problem Formulation

Consider a multispectral image $I \in \mathbb{R}^{C \times H \times W}$, where C is the number of bands, H, W are the height and width of the image respectively. We define a binary segmentation mask $M \in \{0, 1\}^{H \times W}$ over the pixels in the image, where a pixel $M[i, j]$ at row i and column j in the segmentation mask M has a value of 1 if it is predicted to be affected by a landslide, and 0 otherwise.

The Martian landslide detection problem is to develop a segmentation model $S_\theta : \mathbb{R}^{C \times H \times W} \rightarrow \{0, 1\}^{H \times W}$, parameterized by θ , to satisfy the objective

$$\max_{\theta} \mathbb{E}_{(I, M) \sim D} [\sigma_{\text{sim}}(S_\theta(I), M)], \quad (1)$$

where the multispectral image and corresponding segmentation mask pairs (I, M) are sampled from the dataset D , and $\sigma_{\text{sim}}(M_1, M_2) \in \mathbb{R}$ is a similarity metric between two binary segmentation masks M_1 and M_2 . Concretely, the objective aims to maximize the expected similarity between the segmentation mask predicted by the model and the actual segmentation mask, for all examples in the dataset, by performing an optimization over the model parameters θ .

3.2 Joint Embedding Predictive Architectures

3.3 Adapting I-JEPA for Multispectral Image Encoding

3.4 Spatio-Spectral JEPA

3.5 Segmentation Models with Self-Supervised Backbones

4 Experimental Setup

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5 Results and Discussion 256

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6 Conclusion

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Encoder	Segmentation	MIoU	Precision	Recall	F1 Score
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	UNet	0.8286	0.8531	0.8723	0.8624
	UNet++	0.8247	0.8601	0.8795	0.8690
	FPN	0.8246	0.8488	0.8897	0.8687
	MANet	0.8100	0.8448	0.8683	0.8555
	DeepLabv3+	0.8183	0.8576	0.8689	0.8632
	PAN	0.8141	0.8559	0.8693	0.8622
SS-JEPA	UperNet	0.8652	0.8838	0.8859	0.8848
	UNet	0.8521	0.8631	0.8868	0.8748
	UNet++	0.8399	0.8768	0.8596	0.8681
	FPN	0.8265	0.8656	0.8777	0.8716
	MANet	0.8262	0.8691	0.8703	0.8697
	DeepLabv3+	0.8202	0.8579	0.8660	0.8619
	PAN	0.8170	0.8620	0.8587	0.8604

Table 1: Segmentation performance with self-supervised encoder backbones.

Add citations for segmentation heads

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